RIFAT TURJO

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PROFESSIONAL EXPERIENCE

Robotics and Programming Workshop Leader **10/2025 - Present**
TUMO Saar, Saarbrücken

• Delivered workshops in German on the fundamentals of robotics and web development to over 120 participants in C++ and JavaScript
• Improved participant project completion rates by over 95% through individualized feedback and performance evaluation

Embedded Systems Engineer for Sensor Suite Development **03/2023 - 09/2025**
DFKI, Saarbrücken

• Enabled real-time motion tracking across 5 concurrent sensor nodes by designing custom ESP32-based IMU sensors
• Improved motion-tracking accuracy by 40% by implementing advanced digital signal processing algorithms in C++
• Enabled reliable storage and retrieval of sensor measurement data by engineering a data acquisition pipeline with a local server and MySQL database
• Reduced onboarding time for new developers by 3 weeks through writing detailed hardware and firmware documentation

AI Software Developer for LLM Dashboard Development **03/2022 - 03/2023**
DFKI, Saarbrücken

• Built and optimized data acquisition and visualisation pipelines by developing an AI-powered user dashboard in Panel
• Reduced time-to-insight for analytical queries by 66% using an LLM-powered chatbot interface built with Hugging Face
• Assisted with retrieval of analytical results by developing LLM function calling tools in LangChain
• Achieved 95% prediction accuracy in activity-duration estimation by fine-tuning machine learning models in PyTorch
• Reduced data preprocessing time by 80% through cleaning and restructuring datasets using Pandas and NumPy

High School Mathematics and Physics Teacher **01/2019 - 12/2021**
Mastermind English Medium School, Dhaka

• Guided over 200 students for college entrance examinations by preparing lessons and drafting the rationale document for grading exams

EDUCATION

Embedded Systems, Master of Science **10/2021 - 03/2025**
Saarland University, Saarbrücken

Electrical and Electronic Engineering, Bachelor of Science **01/2015 - 08/2018**
BRAC University, Dhaka

TECHNICAL SKILLS

Programming Languages & AI Tools	C, C++, Python, NumPy, Pandas, Verilog, PyTorch, JavaScript, MySQL, Microsoft Copilot, OpenAI Response API, HuggingFace, LangChain
Software Development	GitHub/GitLab, GitHub Actions, CI/CD, Unit Testing/ PyTest, Docker, CMake, Kconfig, FreeRTOS, Zephyr, WSL2, Linux, ROS2
Communication Protocols	SPI, I²C, Wi-Fi, UART, MQTT, Bluetooth, USB, CAN Bus, RS - 232/485, NFC, Ethernet
Hardware Design & Testing	Analog and Digital Circuit Design, Schematic and Layout Design, KiCAD/ Altium Designer, LTSpice/P-spice, Autodesk Fusion 360, Autodesk Inventor Professional, 3D Printing, Oscilloscope, Logic Analyzer, Multimeter, JTAG Debugging, Soft robotics/continuum robots
Embedded Development	Digital Signal Processing, IoT Firmware Development, Espressif ESP32 S3, ARM Cortex-M4F, STM32 Nucleo, Raspberry Pi

LANGUAGES

English: C2 | German: C1 | Bengali: Native

PROJECTS

Zephyr & FreeRTOS on Espressif Development boards — **12/2025 - Present**
Developed cross-platform IoT firmware on an ESP32 S3 development board to resolve hardware interrupts, deadlocks, and priority inversion

Retro Clock Radio — **03/2026 - 06/2026**
Designed a custom PCB clock radio on an ATMEGA328P, integrating FM tuner, RTC, amp, encoder, and display with bare-metal firmware

Master's Thesis — **01/2024 - 01/2025**
Turjo, "SMA Driven Mechanical Metamaterial-based Shape Changing Interfaces." Saarland University, Saarbrücken

Seminar Project — **04/2023- 09/2023**
Turjo, et al. "ParExo: A Parametric Design Tool for Upper-Limb Exoskeletons." HCI Lab, Saarland University, Saarbrücken, Germany

Saarbrücken, 01.09.2026

